

Introduction to Power Query



Data Transformation

- **Objectives:**



- Understand what Power Query is.
- Learn why Power Query is used.
- Understand the ETL process.
- Differentiate between Excel Formulas, Power Query, and DAX.
- Explore where Power Query fits in Power BI and Excel.



1. What is Power Query?

Power Query is a *Data Connection* and *Data Transformation* tool available in both Microsoft Excel and Power BI.

It helps users import data from multiple sources, clean and transform it, and prepare it for analysis without writing complex code.

It follows a low-code/no-code approach, allowing users to perform data preparation using an intuitive graphical interface.

In Simple Words, Power Query is like a washing machine for data.



2. Why Do We Need Power Query?

Real-world data is rarely perfect.
Common problems include:

- Missing values
- Duplicate records
- Extra spaces
- Wrong spellings
- Different date formats
- Blank rows
- Incorrect data types
- Data spread across multiple files



Analyzing dirty data produces incorrect reports and wrong business decisions.

Power Query solves these problems quickly and consistently.



3. Problems Without Power Query

Imagine receiving an Excel file every Monday.
Every week we have to manually:

- Remove duplicates
- Delete blank rows
- Correct spellings
- Change date formats
- Merge columns
- Create new columns

This process takes 20–30 minutes every week.
Over a year: $30 \text{ minutes} \times 52 \text{ weeks} = 26 \text{ hours}$.

Power Query automates these repetitive tasks, reducing them to just a few seconds after the initial setup.



4. What Can Power Query Do?

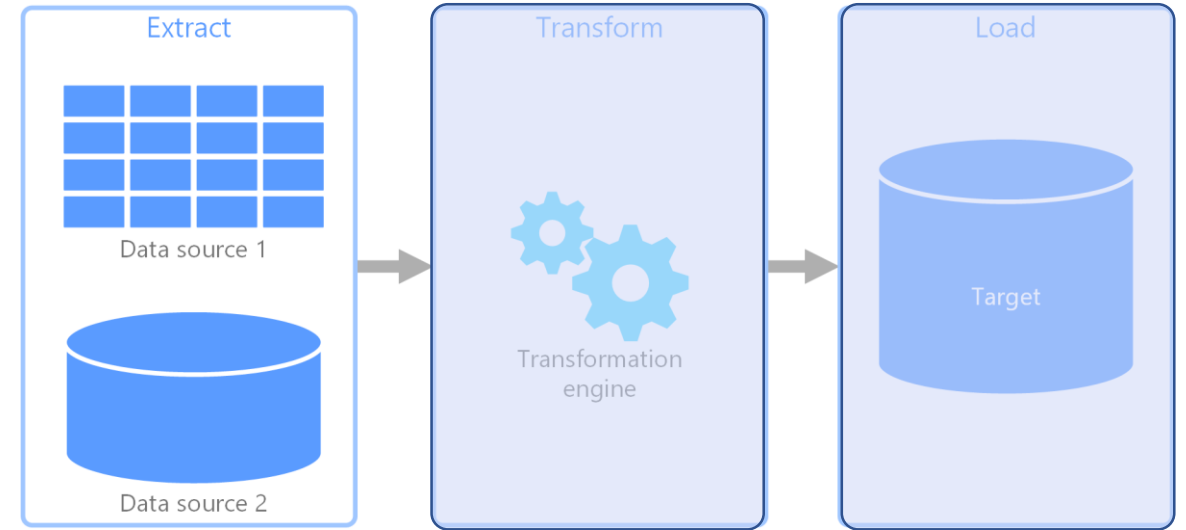
Power Query can:

- Import data from different sources
- Remove duplicate records
- Remove blank rows
- Replace incorrect values
- Change data types
- Split or merge columns
- Pivot and Unpivot tables
- Merge multiple files
- Create calculated columns
- Combine datasets
- Refresh transformed data with one click



5. ETL Concept

Power Query is based on the ETL process.
ETL stands for Extract-Transform-Load



a. Extract:

Extract means collecting data from one or more data sources.

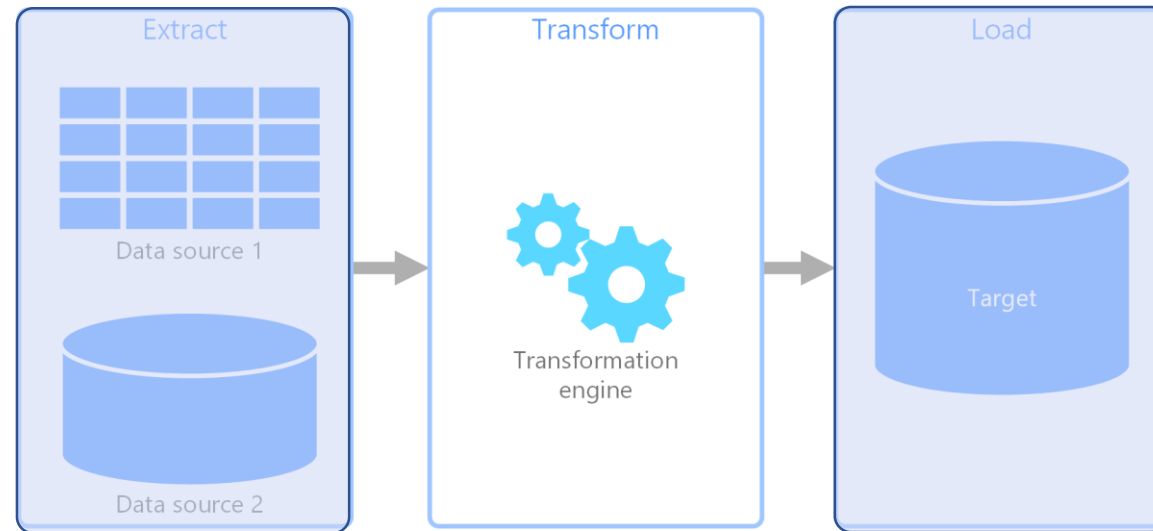
A company stores sales data in Excel and customer data in SQL Server.
Power Query extracts both datasets.



5. ETL Concept

b. Transform:

Transform means cleaning, modifying, and preparing data for analysis. Transformation improves data quality and consistency.



5. ETL Concept

c. Load:

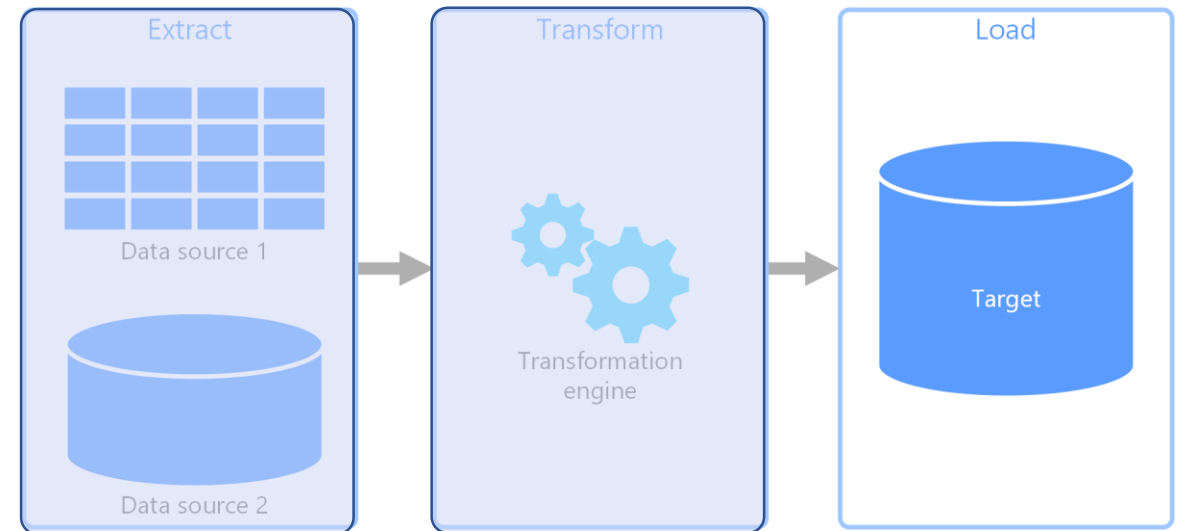
Load means sending the cleaned data to its final destination. Examples:

Power BI Data Model

Excel Table

Warehouse

Dashboard



6. Excel Formulas vs Power Query

Excel Formulas	Power Query
Works inside cells	Works on entire tables
Manual updates	Automatic refresh
Formula for every cell	Transformation for all rows
Difficult for large datasets	Designed for millions of rows
Can become slow	Better performance for data preparation
Easy for calculations	Best for data cleaning

